ment Name: To determine the modulus of rigidity of wire by Maxwell's needle.

APPARATUS:

Maxwell's needle, a wire of suitable length and diameter, stop watch, rigid support, screw guage, spring balance and meter rod.

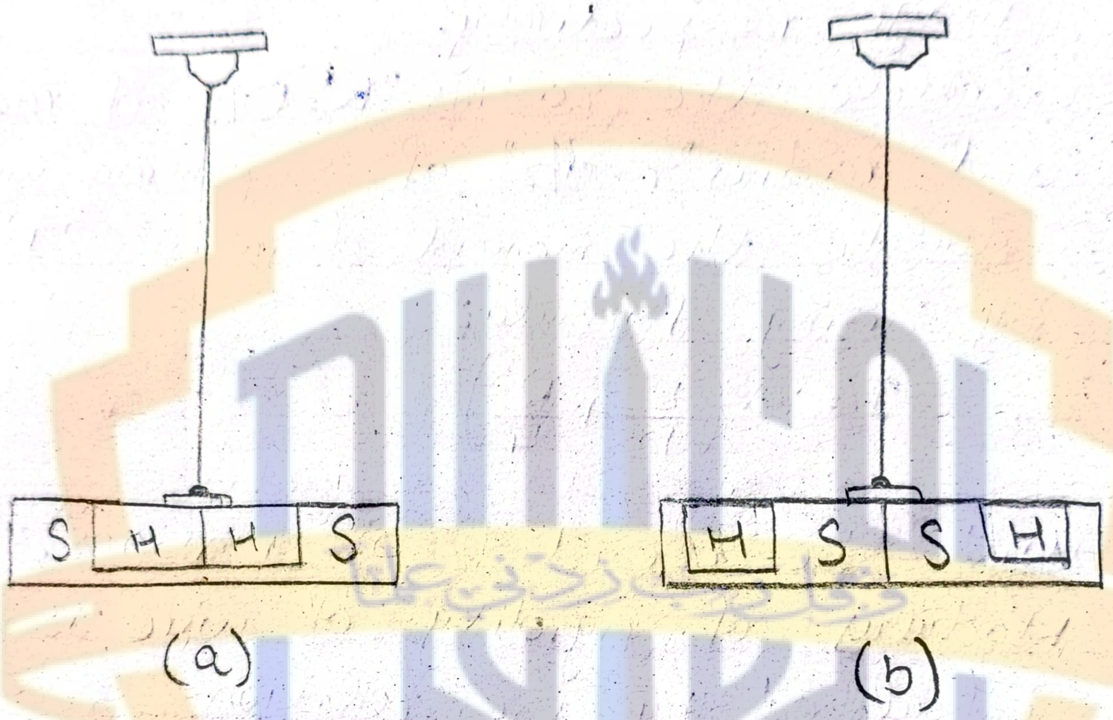
PROCEDURE:

- Take a wire of suitable length and diameter. Suspend the Maxwell's needle from one end of the wire and fix the other end in a rigid support.
- Put the hollow cylinders of Maxwell's needle inward and the solid cylinders outwards symmetrically in the tube.
- Displace one end of the maxwell's tube, and note the position where you released the tube. After oscillating when it comes to same position, 1 vibration is completed.
- Start the stopwatch and find the time for 15 vibrations and repeat it twice. Calculate time period 'T_1' with hollow cylinders inside.

EXPERIMENT

To determine the modulus of rigidity of wire by Maxwell's needle

DIAGRAM:



OBSERVATIONS and CALCULATIONS:

Length of the wire $= l = 87\,cm$

Total length of Maxwell's tube $= 2L = 27\,cm$

Half length of (needle) tube $= L = 13.5\,cm$

Average mass of hollow cylinder $= m_1 = 150\,g$

Average mass of solid cylinder $= m_2 = 189\,g$

ment Name:

- Now repeat the experiment by interchanging both the solid cylinders inward and both hollow cylinders outward. Again find the time for 15 vibrations and repeat it twice. Determine time period 'T_2' with hollow cylinders outsid
- Measure the length 'l' of the wire and also the length 'L' of the Maxwell's needle. Measure the diameter 'd' of the wire with a screw guage. Find out its radius 'r'.
- Determine masses of both the hollow cylinders and find out the average mass 'm_1' of each hollow cylinder. Similarly find out the average mass 'm_2' of each solid cylinder.
- Calculate the modulus of rigidity of a wire by using the formula

$$\eta = \frac{8\pi l(m_2 - m_1)L^2}{(T_1^2 - T_2^2)r^4}$$

Least count of screw guage: 0.001cm

Zero error of screw guage: Nill

Diameter of the wire = d = 0.083cm

Radius of the wire = r = d/2 = 0.0415 cm

Position of Solid Cylinder	Time for 15 Vibrations			Time Period t/15 sec
	1 sec	2 sec	Mean time sec	
Outside	77.4	78	77.7	5.18 = T_1
Inside	70.8	70.2	70.5	4.7 = T_2

Calculations:

$$\eta = \frac{8\pi l(m_2-m_1)L^2}{(T_1^2-T_2^2)r^4}$$

$$= \frac{8\pi(87)(89-50)(13.5)^2}{((5.18)^2-(4.7)^2)(0.0415)^4} = \frac{15541440.01}{2.95\times10^{-5}}$$

$$\eta = 5.27\times10^{11} \text{ dynes cm}^{-2}$$

Result:

$$\eta = 5.27\times10^{11} \text{ dynes cm}^{-2}$$

Actual Value = $6.2 - 8.2\times10^{11}$

Percentage error = $\frac{6.2\times10^{11} - 5.27\times10^{11}}{6.2\times10^{11}}\times100 = 15\%$